6 **(a)** Define electric field at a point.

.....

..... [1]

(b) An isolated conducting sphere in a vacuum has a capacitance of 69 pF.
The charge on the sphere is +83 pC.

(i) On Fig. 6.1, draw field lines to represent the electric field outside the sphere due to the charge on the sphere.

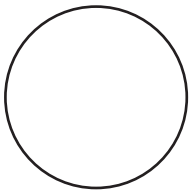


Fig. 6.1

[2]

(ii) Calculate the electric potential at the surface of the sphere.

electric potential = ... V [2]

(iii) Determine the radius of the sphere.

radius = ... m [2]

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(iv) Calculate the electric field strength E at the surface of the sphere. Give a unit with your answer.

E = ... unit ..... [2]

(c) The sphere in **(b)** is discharged by connecting it to earth (0 V) through a resistor of resistance 120 MΩ.

Calculate the time taken for the charge to fall to 26 pC.

time = ... s [2]

[Total: 11]

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